

CLASS
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RATIO & PROPORTION

- 65) Two vessels contain milk & water in the ratio 7:5 and 7:9. If both vessels are mixed in ratio 1:1, find the ratio of milk and water in new mixture?

$$\begin{array}{l} \text{M : W} \\ \text{A } 7 \times 4 : 5 \times 4 = 12 \times 4 \\ \text{B } 7 \times 3 : 9 \times 3 = 16 \times 3 \\ \hline 44 : 47 \end{array}$$

- 66) Three vessels each of ~~10~~¹⁰ litre capacity contain a mixture of milk & water in the ratio 2:1, 3:1 and 3:2. If all the three vessels are emptied into a large vessel, find the ratio of milk & water in new mixture?

$$\begin{array}{l} \text{m : w} \\ \text{A } 2 \times 20 : 1 \times 20 = 3 \times 20 \\ \text{B } 3 \times 15 : 2 \times 15 = 4 \times 15 \\ \text{C } 3 \times 12 : 2 \times 12 = 5 \times 12 \\ \hline 121 : 59 \end{array}$$

- 67) Two vessels A & B contain a mixture of milk & water in the ratio 4:5 and 5:1. ~~In what ratio should quantities of mixture be taken?~~ If both vessels are mixed in the ratio 5:2, find the ratio of milk & water in new mixture.

$$\begin{array}{l} \text{M : W} \\ \text{A } 4 \times 10 : 5 \times 10 = 9 \times 2 = 18 \times 5 \\ \text{B } 5 \times 6 : 1 \times 6 = 6 \times 3 = 18 \times 2 \\ \hline 70 : 56 \Rightarrow 5 : 4 \end{array}$$

$$\begin{array}{l} \text{A} \quad \begin{array}{c} m \\ 1 \times 5 : 2 \times 5 = 3 \times 5 \end{array} \\ \text{B} \quad \begin{array}{c} w \\ 2 \times 3 : 3 \times 3 = 5 \times 3 \\ \hline 11 : 19 \end{array} \end{array}$$

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RATIO & PROPORTION

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- 66) Three vessel each of ~~10~~¹⁰ litre capacity contain a mixture of milk & water in the ratio 2:1, 3:1 and 3:2. if all the three vessels are emptied into a large vessel, find the ratio of milk & water in new mixture?

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- 67) Two vessel A & B contain a mixture of milk & water in the ratio 4:5 and 5:1. ~~In what ratio should quantities of mixture be taken?~~ If both vessel are mixed in the ratio 5:2. find the ratio of milk & water in new mixture.

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(68) If 2 kg of metal of w/c $\frac{1}{3}$ is zinc and rest is copper - 193

is mixed with 3 kg of metal of w/c $\frac{1}{4}$ is zinc and rest is copper

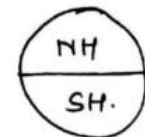
What is the ratio of zinc to copper in new mixture.

$$\begin{array}{ccc} & Z & C \\ A & 1 \times 8 & 2 \times 8 = 3 \times 4 = 12 \times 2 \end{array}$$

$$B. \quad \frac{1 \times 9}{17} : \frac{3 \times 9}{43} = 4 \times 3 = 12 \times 3$$

(69) Ratio of Land : water on earth is 1:2 and ratio of Land : water in northern hemisphere is 2:3. find the ratio of Land : water in southern hemisphere.

$$\begin{array}{ccc} L & W & \\ \text{Earth} & 1 \times 10 : 2 \times 10 = 3 \times 5 = 15 \times 2 \\ \text{NH} & 2 \times 3 : 3 \times 3 = 5 \times 3 = 15 \times 1 \\ \text{SH.} & 4 : 11 \end{array}$$



Earth $\rightarrow 2$
NH $\rightarrow 1$

Earth : NH = 2 : 1

(70) Rs 5600 is to be divided among A, B, C & D in such a way that the ratio of share of A:B is 1:2, B:C is 3:1, C:D is 2:3. find share of (A+B)

$$\begin{array}{cccc} a & : & b & : & c & : & d \\ 1 & : & 2 & & 2 & & 2 \\ 3 & & 3 & : & 1 & & 1 \\ 2 & & 2 & & 2 & : & 3 \\ \hline 6 & : & 12 & : & 4 & : & 6 \\ 3 & : & 6 & : & 2 & : & 3 \end{array}$$

$$\begin{array}{r} 3+6+2+3=14 \\ 14 \text{ — } 5600 \\ 1 \text{ — } 400 \end{array}$$

$$(A+B) = 9 \times 400 = 3600 \text{ Rs.}$$

(71) The ratio of expenditure of A, B and C is 16:12:9 and their total income is 1530. find the share of B's income. if they save 20%, 25% and 40% of their income?

$$\begin{array}{ccc} A & B & C \\ \text{Income} \rightarrow & 20 : & 16 : 15 \\ \text{Exp} \rightarrow & 16 : & 12 : 9 \end{array}$$

$$\begin{array}{l} 51 \rightarrow 1530 \\ 1 \rightarrow 30 \end{array}$$

$$B = 16 \times 30 = 480 \text{ Rs.}$$

$$\begin{array}{l} 20\% = \frac{1}{5} \\ E = 4 \rightarrow 16 \\ 1 \rightarrow 4 \\ 5 \rightarrow 20 \end{array}$$

- 72) The total income of A, B & C is Rs 6060. A spends 80%, B spends 85% and C spends 75%, and the ratio of their saving is 5:6:9. find the income of A. 194

A	B	C	$80\% = \frac{4}{5} \rightarrow 1$	$85\% = \frac{17}{20}$
Saving	5	6	9	$5=3 \rightarrow 6$ $1 \rightarrow 2$ $20 \rightarrow 40$
Income	25	40	36	

$$101 \rightarrow 6060$$

$$1 \rightarrow 60$$

$$A = 25 \times 60 = 1500 \text{ Rs.}$$

- 73) Rs 2366 is divided among 8 men, 10 women & 10 children. Each man gets 25% more than each woman and each woman gets 25% more than each child. find the amount received by each woman.

m	w	c
5	4	4
5	5	4
25	20	16
$\times 8$	$\times 10$	$\times 10$
200	200	160
5	5	4

$$14 \rightarrow 2366$$

$$1 \rightarrow 169$$

$$5 \rightarrow 169 \times 5 = 845$$

$$\text{Each woman} = \frac{845}{10} = 84.5$$

- 74) Rs 500 is divided among A, B, C in such a way that Rs 16 more $\frac{2}{5}$ of A's share, Rs 70 less than $\frac{3}{4}$ of B's share, and Rs 4 less than $\frac{3}{5}$ of C's share are equal. find B's share.

$$A \times \frac{2}{5} + 16 = B \times \frac{3}{4} - 70 = C \times \frac{3}{5} - 4 = 6K$$

$$\text{LCM of } \frac{2}{5}, \frac{3}{4}, \frac{3}{5} = 6$$

$$A = (6K - 16) \times \frac{5}{2} = B = (6K + 70) \times \frac{4}{3} = C = (6K + 4) \times \frac{5}{3}$$

$$A = 15K - 40 \quad B = 8K + \frac{280}{3} \quad C = 10K + \frac{20}{3}$$

$$\therefore 15K - 40 + 8K + \frac{280}{3} + 10K + \frac{20}{3} = 500$$

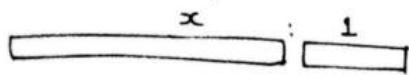
$$23K = 440$$

$$K = \frac{40}{3}$$

$$B = 8 \times \frac{40}{3} + \frac{280}{3}$$

$$= \frac{600}{3} = 200 \text{ Rs}$$

(75) A boy and a girl playing with the pencil. The girl breaks the pencil in two parts and the boy observe that the ratio of length of these two parts is same as the ratio of length of pencil to the larger part. find the ratio in w/c the girl break the pencil.



$$\frac{x}{1} = \frac{x+1}{x}$$

$$x^2 = x+1$$

$$x^2 - x - 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{1 \pm \sqrt{5}}{2}$$

Ratio
a:b नही ले
सकते। 2 variable
दे जायेंगे,
Relation एक
दिता है।
So, x:1 मान
लो।

$$x = \frac{1+\sqrt{5}}{2} \Rightarrow \frac{\sqrt{5}+1}{2}$$

$$\therefore \text{Ratio of two parts} = \sqrt{5}+1 : 2$$

(76) The ratio of income of A & B is 3:2 and ratio of their expenditure is 4:3. if they save Rs 2000 & Rs 900. find their income.

$$\begin{array}{l} \text{A} \quad \text{B} \\ \text{I} \rightarrow 3x : 2x \\ \text{E} \rightarrow 4 : 3 \end{array}$$

$$\frac{3x-2000}{2x-900} = \frac{4}{3}$$

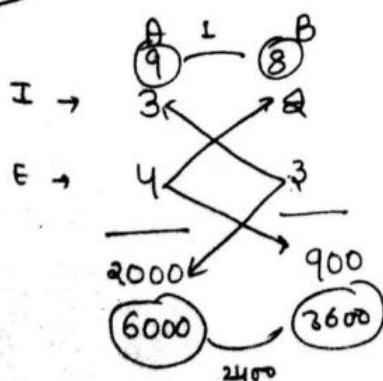
$$9x-6000 = 8x-3600$$

$$x = 2400$$

$$A = 7200 \text{ Rs}$$

$$B = 4800 \text{ Rs}$$

OR



$$1 \rightarrow 2400$$

$$3 \rightarrow 7200 \text{ (A's income)}$$

$$2 \rightarrow 4800 \text{ (B's income)}$$

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- Q22 The ratio of total amount distributed in all the male & female as salary is 6:5 while the ratio of salary of each male & each female is 2:3. find the ratio of no. of male & female?

	m	:	w
Ratio	6	:	5
Each	2	:	3
	<u>3</u>	:	<u>5</u>
	9	:	5

Ans

- Q23 430 is divided among 45 persons such that the ratio of total amt. received by all men, all women & all children are in the ratio 12:15:16. while the ratio of amount received by each man, each woman & each child is 6:5:4. Calculate the no. of man, woman & child and also find the amount received by each of them.

	m	:	w	:	c
Ratio	12	:	15	:	16
Each	6	:	5	:	4
	<u>2</u>	:	<u>3</u>	:	<u>4</u>
	12 men		15 women		20 child

$$9 \rightarrow 45$$

$$1 \rightarrow 5$$

$$12+15+16 = 43 \rightarrow 430$$

$$1 \rightarrow 10$$

$$\text{All man} = 120$$

$$\text{Each man} = \frac{120}{10} = 12 \text{ Rs}$$

$$\text{All women} = 150$$

$$\text{Each woman} = \frac{150}{15} = 10 \text{ Rs}$$

$$\text{Each child} = \frac{160}{20} = 8 \text{ Rs}$$

- Q24 5625 is divided among A, B & C in such a way that A receives $\frac{1}{2}$ the sum of B & C. find the amount received by each. if B receives $\frac{1}{4}$ of (A+C).

$$\frac{A}{B+C} = \frac{1}{2} = \frac{5}{15}$$

$$\frac{B}{A+C} = \frac{1}{4} = \frac{3}{12}$$

$$A : B : C$$

$$5 : 3 : 7$$

$$\frac{A+B}{15} \rightarrow \frac{375}{15} \times 8 = 3000 \text{ Rs}$$

- (80) The ratio of last year income of A, B & C is 3:4:5. while the ratio of their last year income to current year income is 4:5, 2:3 and 3:4. if their total current year income is Rs 98500 find the present income of B+C.

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	A	:	B	:	C	
LYI	3x12	:	4x12	:	5x12	
	(36)		(48)		(60)	
	x9		x24		x20	
	L:C		L:C		L:C	
	4:5		2:3		3:4	
	x9		x24		x20	
	45	:	72	:	80	

LCM of 2,3,4 $\rightarrow$ 12

$45+72+80 = 197 \rightarrow 98500$

$1 \rightarrow \frac{98500}{197} = 500$

$B+C = 152 \times 500$

$= 76000 \text{ Rs. Ans.}$

- (81) one year ago the ratio of income of A & B is 3:5. The ratio of their LYI to CYI is 2:3 and 4:5. if their total CYI is Rs 4300. find their present income individually?

	A		B
Last Year Income	3	:	5
(A)	L:C		(B)
	2:3		L:C
	2 unit $\rightarrow$ 2		4 $\rightarrow$ 5
	1 $\rightarrow$ $\frac{3}{2}$		1 $\rightarrow$ $\frac{5}{4}$
	3 $\rightarrow$ $(\frac{9}{2})$		5 $\rightarrow$ $(\frac{25}{4})$
CYI	A	:	B
	$\frac{9}{2}$	:	$\frac{25}{4}$
	18	:	25
	43 $\rightarrow$ 4300		A $\rightarrow$ 1800 Rs
	1 $\rightarrow$ 100		B $\rightarrow$ 2500 Rs.

OR

	A		B
LYI	3x4	:	5x4
(A)	L:C		(B)
	2:3		L:C
	2 $\rightarrow$ 12		4 $\rightarrow$ 20
	1 $\rightarrow$ 6		1 $\rightarrow$ 5
	3 $\rightarrow$ (18)		5 $\rightarrow$ (25)
	18 : 25		
	A = 1800 Rs		
	B = 2500 Rs		

LCM of 2,4 = 4

- (82) Ratio of income of A, B, C is 3:7:4 and the ratio of their exp. is 4:3:5. if A saves $14\frac{2}{7}\%$ of his income. find the ratio of their saving.

$$\begin{array}{l}
 A : B : C \\
 I \quad 3x : 7x : 4x \\
 E \quad 4y : 3y : 5y \\
 S \quad (3x-4y) : (7x-3y) : (4x-5y)
 \end{array}$$

$$14\frac{2}{7} \cdot \frac{1}{\cdot} = \frac{1}{7} \cdot \frac{S}{I}, E = 6$$

$$\frac{3x}{4y} = \frac{7}{6}$$

$$\frac{x}{y} = \frac{14}{9}$$

$$S = (42-36) : (98-27) : (56-45)$$

$$6 : 71 : 11 \quad \text{Ans}$$

OR

$$\begin{array}{l}
 A \quad B \quad C \quad \text{Ans} \\
 I \quad 7 \quad 196 \quad 112 \quad 14 \\
 E \quad 6 \quad 72 \quad 54 \quad 90 \quad 198 \\
 S \Rightarrow \frac{12}{6} : \frac{142}{71} : \frac{22}{11}
 \end{array}$$

Ako Income & Exp ka ratio 7:6 करना है।

first convert in 1:1 and then in 7:6

- (83) A dog chase a rabbit. The dog takes 6 leaps for every 7 leaps of the rabbit and the distance covered by rabbit in 6 leaps is equal to the distance covered by dog in 5 leaps. find the ratio of their speed.

Dog	Rabbit
6 jump	7 jump
6×6	7×5
$= 36$	$= 35$

Dog	Rabbit
5 jump	= 6 jump

6m 5m

(30)

- (84) A dog takes 7 jumps for every 10 jumps of the lion and a fox takes 12 jumps for every 10 jumps of the lion. And the distance covered by dog in 5 jumps, distance covered by lion in 15 jumps and the distance covered by fox in 20 jumps is equal. find the ratio of their speeds.

Dog	lion	fox
7	10	12
$\times 12$	$\times 4$	$\times 3$
84	40	36

Dog	=	lion	=	fox
5 jump		15 jump		20 jump

12m 4m 3m

(60)

Ratio of the distance covered by two objects in same time is equal to the ratio of their speeds

$D = ST$

85) The price of Gold is directly proportional to square of its weight. A person broke down the Gold in the ratio of 3:2:1 and sold, incurs a loss of Rs 4620. find the initial price of Gold.

initial weight = $3+2+1 = 6$

initial price = $6^2 = 36$

After breaking price = $3^2 + 2^2 + 1^2 = 14$

initial price = $36 \times 210 = 7560 \text{ Rs.}$ Ans.

$36-14 = 22 \text{ part loss}$

$22 \text{ — } 4620$

$1 \text{ — } 210$

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86) A is inversely proportional to the cube of B.

if $A=3$ then $B=2$

but if $A=\frac{8}{9}$ then $B=?$

$A \propto \frac{1}{B^3}$

$A = \frac{24}{B^3}$

$A = \frac{K}{B^3}$

$\frac{8}{9} = \frac{24^3}{B^3}$

$3 = \frac{K}{(2)^3}$

$B^3 = 27$

$B = 3$ Ans.

$K=24$

87) A bag contain 1Rs, 50P & 25P coins and the ratio of no. of coins is 5:7:9. if the total value of all coins is 430 Rs. Then find the no. of 50P coins.

	1Rs	50P	25P
No. of coins	5	7	9
Ratio of value	5Rs	3.50Rs	2.25Rs

$10.75 \text{ Rs — } 430$

$1 \text{ — } \frac{430}{10.75} = 40$

No. of 50P coins = $7 \times 40 = 280$

88) A bag contains 1Rs, 50P & 25P coins and the ratio of their value is 30:11:7. and the total no. of coins are 480. find the no. of 50P coins.

	1Rs	50P	25P
value	30	11	7
coins	30	22	28

$80 \text{ — } 480$

$1 \text{ — } 6$

No. of 50P coins = $22 \times 6 = 132$ By

- 89) find the mean proportion of $\frac{1}{4}$ & $\frac{1}{9}$

$$\sqrt{\frac{1}{4} \times \frac{1}{9}} = \frac{1}{6}$$

$$\text{mean proportion of } a, b = \sqrt{ab}$$

- 90) find the 3rd proportion of 18, 36.

$$\frac{36 \times 36}{18} = 72$$

$$\text{3rd proportion} = \frac{b^2}{a}$$

- 91) find the 4th proportion of 12, 16, 18.

$$\frac{16 \times 18}{12} = 24$$

$$\text{4th proportion of } a, b, c = \frac{bc}{a}$$

- 92) Rs 710 is divided among A, B & C in such a way that A receives Rs 40 more than B, C receives Rs 30 more than A. find the share of A.

A	B	C
$x+40$	x	$x+70$

$$A = 240 \text{ Rs}$$

$$3x + 110 = 710$$

$$3x = 600$$

$$x = 200$$

- 93) The ratio of age of Ram & shyam 5 years ago was 2:3 and the ratio of their age after 5 years would be 3:4 find the sum of their present ages.

	R	S
	$\frac{20}{30}$	$\frac{30}{40}$
Present	$\frac{25}{35}$	$\frac{35}{45}$
	$\frac{3}{4}$	$\frac{4}{5}$
	1 unit	1 unit

$$25 + 35 = 60$$

Ans.

- 94) The ratio of ages of Ram & shyam 5 years ago was 3:1 the ratio of their age after 5 year will be 2:1. find the present age of both.

	R	S
	$\frac{30}{10}$	$\frac{10}{10}$
Present	$\frac{35}{15}$	$\frac{15}{15}$
	$\frac{7}{3}$	$\frac{3}{1}$
	1 unit	1 unit

$$\text{Present age of Ram} = 35$$

$$\text{Present age of Shyam} = 15$$

→ दोनों की age का diff same करने के लिए

(45) The age of Ram is 4 times of his daughter. The age of 201
 Ram was 9 times of her daughter five years ago. - find their
 present ages.

पौत्रों की age का difference
 करने के लिए पहले दोनों
 का diff ले लेते हैं फिर
cross multiply कर देते हैं

$$\begin{array}{rcl}
 R & D & \\
 4x & x & = 8 \\
 \downarrow -5 & \downarrow -5 & \\
 P & D & \\
 4x & x & = 3 \\
 \text{Sum} & \rightarrow & 5 \\
 1 & \rightarrow & 1
 \end{array}$$

Ram = 32 years

daughter = 8 years.

(46) The age of father is 3 times of his son. 5 years before
 the age of son was $\frac{1}{6}$ times of his father. find the
 present age of son. At the time of ^{marriage of} his mother, she was
 5 years younger to his father. find the age of mother.

$$\begin{array}{rcl}
 F & S & \\
 6x & x & = 5 \\
 \downarrow -5 & \downarrow -5 & \\
 P & S & \\
 25 & x & = 2 \\
 \text{Sum} & \rightarrow & 5 \\
 1 & \rightarrow & \frac{5}{2}
 \end{array}$$

Present age of father
 = 25

Present age of son = $5 \times \frac{5}{3} = \frac{25}{3}$

Mother's age $\rightarrow 25 - 5 = 20$

(47) The ratio of age of meena & her mother is 3:8. find the
 ratio of their age after 4 years. if after 10 years their age
 diff will be 35 years.

$$\begin{array}{rcl}
 M & \text{Mother} & \\
 3 & 8 & \\
 \downarrow +4 & \downarrow +4 & \\
 7 & 12 & \\
 \downarrow +10 & \downarrow +10 & \\
 17 & 22 & \\
 \text{Diff} & \rightarrow & 5 \\
 1 & \rightarrow & 7
 \end{array}$$